Addition of rational numbers

A common denominator, then the integer sign rules — nothing else is new.

LESSON 39–40 | 2 × 40 MIN | MATEMATIKA 7, §16 | S8 7.2

LESSON CLOCK

15' | 25' | 25' | 20' | 5'

Same denominator	15 min
Different denominators	25 min
With signs	25 min
Mixed numbers	20 min
Homework	5 min

LEARNING OBJECTIVES

- Add fractions with the same and with different denominators.
- Add positive and negative fractions using the sign rules.
- Add mixed numbers and decimals.
- Use the commutative and associative laws to simplify a sum.

TEXTBOOK

National	pp. 101–107
Cambridge	Stage 8, pp. 70–76
Workbook	Exercise 7.2

01 Explanation

Same denominator

$$\frac{a}{c} + \frac{b}{c} = \frac{a+b}{c}$$

Add the numerators and keep the denominator; then simplify.

SUM	WORKING	ANSWER
$\frac{2}{7} + \frac{3}{7}$	$\frac{5}{7}$	$\frac{5}{7}$
$\frac{3}{8} + \frac{1}{8}$	$\frac{4}{8}$	$\frac{1}{2}$
$-\frac{5}{9} + \frac{2}{9}$	$\frac{-3}{9}$	$-\frac{1}{3}$

DENOMINATORS ARE NOT ADDED

$\frac{2}{7} + \frac{3}{7}$ is $\frac{5}{7}$, not $\frac{5}{14}$. Thinking of sevenths as a unit — two sevenths plus three sevenths — makes the rule obvious.

Different denominators

STEP	WHAT TO DO
1	find the LCM of the denominators
2	rewrite each fraction with that denominator
3	add the numerators
4	simplify

Example. $\frac{3}{4} + \frac{5}{6}$. LCM of 4 and 6 is 12:

$$\frac{9}{12} + \frac{10}{12} = \frac{19}{12} = 1\frac{7}{12}$$

USE THE LCM, NOT THE PRODUCT

$4 \times 6 = 24$ also works, but leaves $\frac{38}{24}$ to simplify. The LCM gives the shortest route and the least cancelling.

With signs

Once the denominators agree, the numerators are added exactly as integers were in the last chapter.

SUM	COMMON DENOMINATOR	ANSWER
$-\frac{1}{2} + \frac{1}{3}$	$-\frac{3}{6} + \frac{2}{6}$	$-\frac{1}{6}$
$-\frac{3}{4} + (-\frac{1}{6})$	$-\frac{9}{12} - \frac{2}{12}$	$-\frac{11}{12}$
$\frac{5}{8} + (-\frac{5}{8})$	$\frac{0}{8}$	0

TWO CHAPTERS, ONE CALCULATION

Chapter II supplied the sign rules; this chapter supplies the common denominator. Every question here is one of each, done in that order.

Mixed numbers

Two routes: convert to improper fractions, or add whole parts and fractional parts separately.

SUM	ROUTE A	ROUTE B
$2\frac{1}{3} + 1\frac{1}{2}$	$\frac{7}{3} + \frac{3}{2} = \frac{23}{6}$	$3 + \frac{5}{6}$

Both give $3\frac{5}{6}$. Route B is quicker when the whole parts are large; Route A is safer when signs are involved.

WITH NEGATIVE MIXED NUMBERS, CONVERT FIRST

$-2\frac{1}{3}$ means $-(2 + \frac{1}{3}) = -\frac{7}{3}$, not $-2 + \frac{1}{3}$. Route A removes the ambiguity entirely.

02 Worked examples

EXAMPLE 1 **Compute $\frac{3}{4} + \frac{5}{6}$.**

1. LCM of *4* and *6* is *12*.
2. $\frac{9}{12} + \frac{10}{12}$
3. $= \frac{19}{12}$
4. $= 1\,\frac{7}{12}$

ANSWER

$1\,\frac{7}{12}$

EXAMPLE 2 **Compute $-\frac{1}{2}+\frac{1}{3}$ and $-\frac{3}{4}+\left(-\frac{1}{6}\right)$.**

1. $-\frac{3}{6}+\frac{2}{6}=-\frac{1}{6}$

   Different signs.

2. LCM of *4* and *6* is *12*.

3. $-\frac{9}{12}-\frac{2}{12}$

   Same signs.

4. $=-\frac{11}{12}$

ANSWER

$-\frac{1}{6}$ and $-\frac{11}{12}$

EXAMPLE 3 **Compute $2\dfrac{1}{3} + 1\dfrac{1}{2}$.**

1. Whole parts: $2 + 1 = 3$.

   Route B.

2. Fractions: $\dfrac{2}{6} + \dfrac{3}{6} = \dfrac{5}{6}$

3. $3 + \dfrac{5}{6}$

4. $= 3\dfrac{5}{6}$

ANSWER

$3\dfrac{5}{6}$

03 Interactive model

Cut two paper strips into quarters and sixths; the class sees that twelfths are the smallest pieces both can be made from, and the LCM stops being a rule.

Finding the common denominator

DENOMINATOR	FACTORISED	EXTRA FACTOR NEEDED
6	$2 \cdot 3$	4
8	2^3	3

LOWEST COMMON DENOMINATOR

$$24$$

$$\frac{20}{24} + \frac{9}{24} = \frac{29}{24}.$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Common denominator	Umumiy maxraj	Общий знаменатель
Lowest common denominator	Eng kichik umumiy maxraj	Наименьший общий знаменатель
To simplify	Qisqartirish	Сократить
Mixed number	Aralash son	Смешанное число
Improper fraction	Noto‘g‘ri kasr	Неправильная дробь
Sum	Yig‘indi	Сумма
Like denominators	Bir xil maxrajlar	Одинаковые знаменатели
LCM	EKUK	НОК

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\frac{2}{7} + \frac{3}{7}$ equals:

- $\frac{5}{14}$
- $\frac{5}{7}$
- $\frac{6}{7}$
- $\frac{6}{49}$

For different denominators, first find:

- the product
- the LCM
- the HCF
- the sum

$\frac{3}{4} + \frac{5}{6}$ equals:

- $\frac{8}{10}$
- $1\frac{7}{12}$
- $\frac{19}{24}$
- 2

$-\frac{1}{2} + \frac{1}{3}$ equals:

- $\frac{1}{6}$
- $-\frac{1}{6}$
- $-\frac{5}{6}$
- $\frac{5}{6}$

$-2\frac{1}{3}$ as an improper fraction:

- $-\frac{7}{3}$
- $-\frac{5}{3}$
- $\frac{7}{3}$
- $-\frac{1}{3}$

$\frac{5}{8} + (-\frac{5}{8})$ equals:

- $\frac{10}{8}$
- 0
- $\frac{5}{4}$
- $-\frac{5}{8}$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $\frac{2}{7} + \frac{3}{7}$

ANSWER $\frac{5}{7}$

2. $\frac{3}{8} + \frac{1}{8}$

ANSWER $\frac{1}{2}$

3. $\frac{1}{2} + \frac{1}{4}$

ANSWER $\frac{3}{4}$

4. $\frac{1}{3} + \frac{1}{6}$

ANSWER $\frac{1}{2}$

5. $-\frac{5}{9} + \frac{2}{9}$

ANSWER $-\frac{1}{3}$

6. $\frac{5}{8} + (-\frac{5}{8})$

ANSWER 0

7. $0.4 + 0.35$

ANSWER 0.75

Medium · the standard question

7 PROBLEMS

1. $\frac{3}{4} + \frac{5}{6}$

ANSWER $1\frac{7}{12}$

2. $-\frac{1}{2} + \frac{1}{3}$

ANSWER $-\frac{1}{6}$

3. $-\frac{3}{4} + (-\frac{1}{6})$

ANSWER $-\frac{11}{12}$

4. $2\frac{1}{3} + 1\frac{1}{2}$

ANSWER $3\frac{5}{6}$

5. $\frac{2}{3} + \frac{3}{5}$

ANSWER $1\frac{4}{15}$

6. $-\frac{5}{6} + \frac{3}{4}$

ANSWER $-\frac{1}{12}$

7. $1\frac{1}{4} + (-\frac{1}{2})$

ANSWER $\frac{3}{4}$

Hard · stretch and reason

7 PROBLEMS

1. $\frac{1}{2} + \frac{1}{3} + \frac{1}{6}$

ANSWER 1

2. $-\frac{2}{3} + \frac{3}{4} - \frac{1}{2}$

ANSWER $-\frac{5}{12}$

3. $-1\frac{2}{5} + 2\frac{3}{10}$

ANSWER $\frac{9}{10}$

4. $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16}$

ANSWER $\frac{15}{16}$

5. Find x: $x + \frac{2}{3} = \frac{1}{6}$

ANSWER $-\frac{1}{2}$

6. $0.25 + \frac{1}{3}$

ANSWER $\frac{7}{12}$

7. The mean of $\frac{1}{2}$ and $-\frac{1}{6}$

ANSWER $\frac{1}{6}$

07 Homework

Homework — 5 tasks

Find the LCM before writing anything, and simplify every answer.

1. Compute $\frac{2}{5} + \frac{1}{3}$ and $\frac{5}{6} + \frac{3}{8}$.
2. Compute $-\frac{2}{3} + \frac{1}{4}$.
3. Compute $-\frac{1}{5} + (-\frac{3}{10})$.
4. Compute $3\frac{1}{4} + 2\frac{2}{3}$.
5. Find x if $x + \frac{3}{4} = \frac{1}{2}$.